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AI-assisted speaking affordances, flow experience, and willingness to communicate: A moderated mediation model

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ABSTRACT

Existing research has widely examined the effects of AI-assisted speaking training on language performance and learning outcomes; however, the experiential mechanisms through which AI-assisted speaking training affordances (AISTA) shape learners' communication-oriented behaviors remain underexplored. In particular, limited empirical evidence explains how such affordances are translated into learners' willingness to communicate (WTC) via flow experience (FE). Drawing on ecological affordance theory and flow theory, the present study conceptualizes AISTA as a core construct and proposes a moderated mediation model in which AISTA influences WTC through FE, with learner agency (LA) capturing individual differences in AI-supported speaking contexts.

Survey data were collected from 247 Chinese EFL university students following a four-week AI-assisted speaking practice and analyzed using structural equation modeling (SEM) with latent main effects and interaction terms. Bias-corrected bootstrapping with 5000 resamples was employed to test mediation and moderation effects. The results showed that AISTA was positively associated with FE, and FE was positively associated with WTC. Flow experience emerged as a key mediator linking AISTA to WTC. Moreover, learner agency significantly moderated the experiential pathway, yielding a pattern of conditional partial mediation: the indirect effect of AISTA on WTC via FE was significant for learners with moderate to high levels of agency but not for those with low agency.

By elucidating the affordance–experience–behavior mechanism underlying AI-assisted speaking training, this study extends ecological affordance theory and flow theory to AI-supported second language learning contexts and highlights learner agency as a critical condition for translating technological affordances into communicative motivation.

1. Research background

The rapid advancement of artificial intelligence (AI) has profoundly transformed educational practices, and its integration into foreign language learning has attracted growing scholarly attention. In recent years, generative AI-based speaking tools have proliferated, offering learners more flexible opportunities for oral practice and immersive simulated communication environments. Recent systematic review (e.g., Albedah, 2025) indicates that the application of AI in language education has evolved beyond early computer-assisted instruction to encompass multidimensional functions such as generative feedback, semantic evaluation, and context-based conversational interaction. These technologies have helped alleviate long-standing challenges in traditional

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classrooms—namely, language anxiety, limited speaking opportunities, and delayed feedback—thereby opening new pathways for oral language instruction (Liu, Darvin, & Ma, 2025; Peng & Liang, 2025).

However, AI technologies do not automatically translate into learning success. Prior studies have shown that when human-AI interactions are poorly aligned, learners may experience frustration or even motivational decline (Kim & Su, 2024). Consequently, the educational value of AI lies not in the technology itself but in how learners perceive and mobilize the opportunities it affords. In this regard, AI-assisted speaking training affordances (AISTA) emerge as a critical analytical construct, emphasizing learners' perception and activation of action possibilities within AI-mediated environments rather than the technical functionalities per se (Song, 2021; Van Lier, 2004).

At the same time, research attention must also be directed to the dynamic psychological dimension of learning. Flow experience (FE), conceptualized by Csikszentmihalyi (1990) as a state of intense immersion, concentration, and enjoyment, has been extensively validated as a key psychological condition that fosters learning motivation and engagement (Egbert, 2004; Engeser & Rheinberg, 2008; Zuniga, 2023). Within AI-mediated interaction, an important yet underexplored question concerns whether AISTA can consistently elicit flow and, in turn, enhance learners' willingness to communicate (WTC) in a second language. WTC is widely regarded as the most immediate antecedent of language output, determined not only by linguistic competence and anxiety but also by the learner's situational experience (Dewaele & Dewaele, 2018; Kirkpatrick et al., 2024).

Recent findings further suggest that learner autonomy and positive emotional experiences play increasingly significant roles in promoting WTC (Sato, 2025). As Tarp (2025) argues, learners' psychological experiences in AI-mediated contexts may foster WTC through the joint activation of positive emotions and agentic behaviors. Taken together, these findings indicate that flow experience likely serves as the crucial psychological pathway linking AISTA and WTC.

Moreover, individual differences among learners should not be overlooked. Prior research has demonstrated that learners exposed to the same AI environment may experience markedly different levels of engagement and learning outcomes (Lee & Lee, 2024; Peng & Liang, 2025). Learner agency (LA) has been identified as a key factor underlying such variability, as it shapes how individuals actively construct, regulate, and transform their learning experiences (Reeve & Tseng, 2011; Song, 2021). Yet, the precise role of LA within the technology-psychology-behavior chain remains insufficiently examined, leaving open the question of how learners perceive and mobilize technological resources in AI-mediated contexts. In other words, current research calls for a comprehensive framework that integrates environmental affordances, psychological experience, and learning behavior, thereby revealing how learners activate latent action possibilities in complex technology-enhanced environments.

Building on the above issues, this study adopts Affordance Theory as its overarching theoretical framework. Gibson (1979) first introduced the concept of affordances to refer to the action possibilities that the environment offers to an actor, emphasizing that affordances are not objective functions inherent in the environment but relational outcomes emerging from the interaction between an individual's capabilities and environmental properties (pp. 127–129). Later, Van Lier (2004) introduced this concept into the field of language learning, arguing that linguistic affordances do not stem from technology itself but from learners' ability to perceive, select, activate, and utilize the linguistic resources offered by their environment. In other words, the same educational technology may not afford identical action possibilities to all learners. This theoretical stance provides the rationale for positioning the AI-assisted speaking training affordances (AISTA) as the core construct of the present study.

Within the context of generative AI-based speaking training, the system's immediate feedback, adaptive adjustments, and authentic situational simulations constitute a set of potential affordances. Only when learners actively explore and activate these affordances can effective learning experiences occur (Kordt, 2016; Song, 2021). Integrating AISTA, FE, LA, and WTC within a unified ecological framework thus offers a coherent explanation for the dynamic technology → psychological experience → behavioral motivation chain. From this theoretical standpoint, the limitations of existing research also become more apparent.

Taken together, the existing literature presents three gaps. First, prior work has focused primarily on technology acceptance and learning outcomes, while the potential pathway from AISTA to WTC via FE has received little attention. Second, individual differences

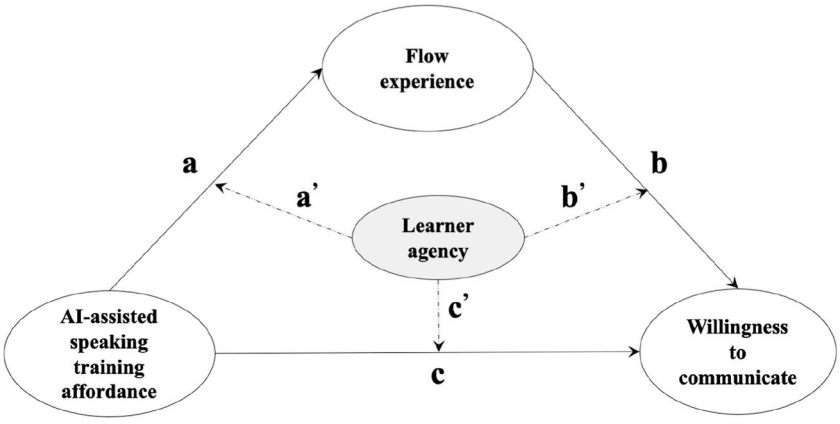


Fig. 1. Solid lines represent the hypothesized main effects, while dashed lines indicate the moderating effects of learner agency (LA). All paths are conceptual and based on the proposed research framework.

have not been adequately addressed, and the moderating role of LA remains underexplored. Third, there is a lack of integrative models that explain the dynamic relationship among technology, psychological experience, and communicative behavior. To address these gaps, the present study draws on ecological affordance theory, flow theory, and the WTC framework to construct a mediation model with LA as a central moderator. Specifically, it examines (1) whether the relationship between AISTA and WTC is mediated by FE, and (2) whether LA moderates the key paths within this process (see Fig. 1).

2. Theoretical framework

2.1. Ecological affordance theory and the definition of AISTA

As noted above, according to Ecological Affordance Theory, learning opportunities are not automatically generated by technology itself but rather emerge from the dynamic interaction between environmental attributes and learners' capabilities (Gibson, 1979). Van Lier (2004) further emphasized that language learning occurs only when learners notice, recognize, and mobilize the linguistic resources available in their environment. Drawing on this perspective, the present study defines AI-assisted speaking training affordances (AISTA) as the action possibilities that learners perceive and are able to activate when interacting with generative AI tools. This ecological view provides the theoretical rationale for positioning AISTA at the core of the proposed model.

2.2. Theoretical foundations and definitions of core constructs

Within the overarching ecological framework, the four core variables examined in this study play distinct yet interrelated theoretical roles in the proposed model. AISTA, as an external environmental input, does not merely refer to the technological properties of AI tools; rather, it reflects learners' overall perception of features such as controllability, interactivity, immediacy of feedback, and adaptive adjustment. These perceived attributes collectively determine whether learners can effectively activate the available affordances (Kearney et al., 2019; McMillan & Hwang, 2002).

Correspondingly, FE, derived from Csikszentmihalyi's (1990) Flow Theory, emphasizes immersion, clarity of goals, and the balance between challenge and skill. FE represents a critical psychological state that sustains learning engagement and motivation (Egbert, 2004; Zuniga, 2023).

LA, grounded in Reeve and Tseng's (2011) concept of "agentic engagement," refers to learners' capacity to actively shape their learning conditions through questioning, seeking feedback, and regulating strategies, thereby influencing the extent to which they can activate AISTA (Lai & Lee, 2024).

Finally, WTC, following the pyramid model proposed by MacIntyre et al. (1998), is regarded as the most immediate motivational antecedent of language output. It is jointly influenced by situational confidence, emotional experience, and perceived communicative safety.

2.3. The path mechanism of affordance, experience, and motivation

Integrating the above constructs, a clear "affordance-experience-motivation" chain can be established. Learners' perception of AISTA constitutes the environmental foundation that triggers FE. When the AI environment is perceived as more controllable, responsive, and meaningful, learners are more likely to experience immersion and enjoyment during interaction. The immediacy of feedback, interactivity, and contextual authenticity offered by AISTA create a balanced challenge-skill environment that facilitates the emergence of flow (Csikszentmihalyi, 1990; Egbert, 2004). Such positive experiences not only enhance learning enjoyment and self-efficacy but also shift language practice from passive completion to active exploration, thereby stimulating sustained learning motivation and willingness to communicate (Kirkpatrick et al., 2024; Zuniga, 2023).

From the perspective of WTC theory, the core driving force of language output arises from the accumulation of positive situational experiences (MacIntyre et al., 1998). In AI-supported speaking environments, fluent, enjoyable, and psychologically safe interactions can substantially reduce communication anxiety and strengthen learners' sense of security, consequently increasing their willingness to express themselves in authentic communicative contexts (Alrabai, 2022; Dewaele & Dewaele, 2018). The AISTA → FE → WTC pathway thus reveals the underlying psychological mechanism of technology-enhanced learning and offers a theoretical anchor for understanding how motivation emerges in human-AI symbiotic environments.

Within this process, LA plays a critical conditional role. Learners with high levels of agency are better able to notice and utilize the multiple affordances offered by AI, actively engage with challenges and supports, and are therefore more likely to achieve flow and develop stronger communicative motivation. In contrast, those with low agency may fall into an affordance blindness, failing to fully activate the potential of their learning environment. Overall, this logic chain aligns with the core propositions of positive psychology, technology-assisted learning, and WTC theory, forming the theoretical foundation of the proposed research model.

In summary, the "affordance-experience-motivation" framework elucidates the intrinsic relationships between technological context, psychological experience, and communicative motivation, while underscoring the pivotal role of learner agency in mediating these dynamics. Based on this theoretical logic, the following section specifies the corresponding research hypotheses for empirical testing.

2.4. Research questions and hypotheses

Building on the theoretical framework outlined above, this study is situated in the context of AI-assisted speaking training and aims to examine how learners' perceptions of technological affordances influence their communication-oriented behavioral tendencies through psychological experience mechanisms. In addition, the study explores the moderating role of learner agency in shaping these relationships. Specifically, the study seeks to address the following research questions:

RQ1: Are learners' perceived AI-assisted speaking training affordances (AISTA) significantly associated with their flow experience (FE) and willingness to communicate (WTC)?
RQ2: Does flow experience (FE) mediate the relationship between AISTA and WTC, thereby constituting an affordance-experience-communication pathway?
RQ3: Does learner agency (LA) moderate the relationships between AISTA and FE, between FE and WTC, and between AISTA and WTC?
RQ4: Does the indirect effect of AISTA on WTC via flow experience vary across different levels of learner agency?

Based on these research questions and the preceding theoretical integration and logical reasoning, the following hypotheses are proposed. The theoretical model of the study is presented in Fig. 1.

H1. AISTA is positively associated with FE (Path a, see Fig. 1).

H2. FE is positively associated with WTC (Path b, see Fig. 1).

H3. AISTA is positively associated with WTC (Path c, see Fig. 1).

H4. Flow experience (FE) mediates the relationship between AISTA and WTC, such that the effect of AISTA on WTC is transmitted through an indirect pathway via FE (Path a → b, see Fig. 1).

H5. H7: Learner agency (LA) moderates multiple pathways in the proposed model, specifically by moderating the relationships between AISTA and FE (Path a′, see Fig. 1), between FE and WTC (Path b′, see Fig. 1), and between AISTA and WTC (Path c′, see Fig. 1).

3. Research methods

3.1. Participants and research context

This study recruited 278 non-English-major undergraduates from a comprehensive university in China. A total of 261 students completed all experimental sessions and the final survey, resulting in 247 valid responses (valid response rate = 94.6 %). Among the valid sample, participants ranged in age from 18 to 22 years (M = 19.6, SD = 0.9), with females accounting for 67.2 %. All participants had completed at least three semesters of college English courses within the past year, had passed the College English Test Band 4 (CET-4), and received training in AI-assisted speaking practice prior to the study. To control for external variables, all participants engaged in the experiment under comparable instructional arrangements and technological conditions.

The study adopted Tencent Yuanbao, a generative AI–based speaking practice tool developed by Tencent, as the technological platform. The system supports speech-enabled interaction and automated feedback by analyzing learners' spoken input in real time. During interaction, the tool provides individualized feedback on grammar, vocabulary, and pronunciation, and adjusts task difficulty and feedback content dynamically to sustain learner engagement and interactional continuity.

For the 10-min AI speaking task, this study adopted Scrivener's (2011) recommendation that delayed error correction facilitates

Table 1
Research variables and measurement instruments.

Variable	Scale Sources and Adaptation Basis	Main Contents	Example Items
AISTA	Adapted from the Perceived Interactivity Scale (McMillan & Hwang, 2002) and the Mobile Learning Affordance Scale (Kearney et al., 2019)	Learner-perceived action possibilities in AI interaction: user control, bidirectional communication, synchronous feedback, personalized difficulty, authentic context, collaborative practice	"When practicing speaking with AI, I can actively control the direction and pace of the conversation, which reflects my ability to shape the interaction."
FE	Adapted from the Flow Short Scale (Engeser & Rheinberg, 2008; Rheinberg et al., 2003), contextualized to language learning (Egbert, 2004)	Concentrated involvement, altered sense of time, fluency of expression, goal clarity, sense of control (including one reverse-coded item)	"When practicing speaking with AI, I often lose track of time."
WTC	Based on Peng and Woodrow's (2010) Chinese EFL Classroom WTC Scale and MacIntyre et al.'s (1998) situational WTC framework	Willingness to initiate speaking, classroom participation, spontaneous interaction, cross-situational use	"In conversations with AI, I am willing to take the initiative and try to expand my responses."
LA	Adapted from the Agentic Engagement Scale (Reeve & Tseng, 2011)	Active involvement, guiding interaction, strategy adjustment, transfer to real contexts	"I take the initiative to ask AI more challenging questions to push the conversation forward."

spoken fluency. Accordingly, the frequency of AI feedback was controlled at approximately five to eight corrections within a 10-min conversation, in order to avoid excessive interruptions that might disrupt communicative coherence.

To ensure consistency in human-AI interaction, participants were provided with standardized AI prompts specifying the dialogue setup, feedback frequency, difficulty adjustment, and feedback format. For example, students were instructed to initiate each session with a standardized command such as: "*Please engage in a 10-min English conversation with me on the topic of ____. Provide light feedback every one to 2 min, and at the end, categorize your feedback into grammar, vocabulary, expression, pronunciation, and content logic.*"

Following this, students processed AI feedback using a "record-organize-synthesize" approach: during interaction, they noted key points without interrupting the flow of conversation; after the session, they organized the feedback into categories and summarized frequent errors as well as strategies for improvement.

This training procedure preserved the pedagogical value of AI's immediate feedback while minimizing interference with fluency. It enabled learners to maintain a balance between accuracy and communicative flow and, through systematic reflection and strategy generation, to more effectively activate and utilize AISTA.

3.2. Instruments

Four core variables were measured in this study, all adapted from well-established scales with contextual modifications to fit the AI-assisted speaking practice setting. The items were designed to capture learners' perceived action possibilities in interaction with AI environments rather than the technical functions of the tool itself (see Table 1). All items were translated into Chinese and refined through bilingual expert review and back-translation procedures.

The AISTA and FE constructs were each measured with six items, while WTC and LA were each measured with five items. All scales used a 7-point Likert format (1 = "strongly disagree," 7 = "strongly agree"), with higher scores indicating higher levels of the construct. To ensure reliability and validity, the instruments were reviewed by experts and piloted with a small sample before the formal survey. Exploratory factor analysis confirmed the unidimensional structure of the constructs, with all item loadings exceeding 0.60. The Cronbach's alpha values for the four scales ranged from 0.71 to 0.84, indicating satisfactory internal consistency.

For flow experience (FE), the English version of the Flow Short Scale was used, based on the original Flow State Scale developed by Rheinberg et al. (2003). The scale was adapted to the AI-assisted speaking context to capture learners' momentary immersion, perceived control, and altered sense of time during human-AI interaction.

3.3. Data collection procedure

The experimental period of this study was set to four weeks, based on two major considerations. First, existing research on AI-assisted speaking training commonly adopts short-term designs ranging from three to six weeks, aiming to capture the immediate psychological and behavioral effects of technology use (Tai & Chen, 2024; Yang & Li, 2024). Both FE and WTC are highly state-dependent variables whose accumulative effects can be effectively observed within shorter intervention cycles. Accordingly, a four-week duration was deemed sufficient to yield stable and representative psychological and behavioral indicators.

Second, participants engaged in two AI-assisted speaking sessions per week. While a regular class period lasted approximately 40 min, the AI-mediated speaking task used for the intervention and subsequent evaluation consisted of a standardized 10-min conversation, followed by brief note-taking and post-task organization of AI feedback. This design ensured sufficient interactional frequency for learners to become familiar with the AI's feedback patterns and develop stable experiential trajectories, while avoiding fatigue effects or participant attrition that might arise from longer interventions.

With regard to extracurricular use, students were not required to conduct additional AI-based practice outside the scheduled sessions, thereby minimizing potential noise caused by variations in practice frequency. To ensure workload consistency, all participants completed the eight assigned AI speaking tasks within designated time windows supervised by the research team. Although extracurricular use was not mandatory, students were permitted to engage in voluntary, low-stakes interactions with AI in their daily communication, maintaining a naturalistic learning ecology.

Regarding topic design, each session included one assigned topic and one open-ended topic. The assigned topics, developed by the research team, covered a range of communicative contexts such as campus life (e.g., "*Differences in study habits at university*"), interpersonal communication (e.g., "*Managing roommate conflicts*"), future planning (e.g., "*Career aspirations and expectations*"), and social issues (e.g., "*The impact of AI on future work*"). These topics ensured moderate challenge in terms of linguistic load, cognitive complexity, and interactional depth. The open-ended topics allowed students to initiate conversations based on their personal interests and communicative needs, enhancing authenticity and autonomy. This balance between structured and self-directed tasks maintained consistency in training while preserving individual variation in interactional patterns. Although the speaking practice was implemented over four weeks, the present analysis focuses on cross-sectional survey data collected after the intervention.

3.4. Data analysis

Structural equation modeling (SEM) was used to test the proposed model and hypotheses in two steps. First, the measurement model was validated to assess the reliability and validity of the latent constructs. Second, the structural model was analyzed to evaluate the path coefficients for H1-H7 and the overall model fit.

After the measurement model was validated, model parameters were estimated using the maximum likelihood (ML) method. Mediation and moderation effects were examined with bias-corrected bootstrapping (5000 resamples) to generate confidence

intervals. Because the model incorporated moderation, interaction terms were introduced into the SEM. Specifically, the interaction between AISTA and LA was added to the FE equation to test H5, the interaction between FE and LA was added to the WTC equation to test H6, and the interaction between AISTA and LA was used to directly predict WTC to test H7.

All continuous variables were mean-centered before constructing interaction terms to reduce the risk of multicollinearity. For significant interactions, simple slope analyses were conducted at high (+1 SD) and low (−1 SD) levels of LA, and slope graphs were generated to illustrate the interaction patterns.

In addition, mediation through FE in the "AISTA → WTC" pathway (H4) was tested, and the analysis further examined whether this indirect effect varied by levels of LA, constituting a moderated mediation effect.

3.5. Control of common method bias

Since all variables were derived from self-reported questionnaire data collected at a single time point, several procedural and statistical remedies were employed to minimize potential common method bias (CMB). Procedurally, participants completed the survey anonymously, with randomized item order and inclusion of reverse-coded items to reduce response pattern bias.

Statistically, an unrotated single-factor test indicated that the first principal factor accounted for 38.97 % of the total variance, which is below the commonly accepted threshold of 50 %. Furthermore, a single-method-factor confirmatory factor analysis (CFA) and marker variable technique were conducted for cross-validation. The inclusion of a method factor did not significantly improve model fit ($\Delta\chi^2 = \text{ns}$), suggesting that common method bias was unlikely to threaten the validity of the main findings (Podsakoff et al., 2003).

4. Results

4.1. Descriptive statistics and correlation analysis

Table 2 reports the reliability and validity of the constructs, along with inter-construct correlations. The results indicate that AISTA, FE, WTC, and LA were all significantly and positively correlated ($r = 0.255–0.724$, $p < .01$), providing preliminary support for subsequent structural equation modeling analyses.

4.2. Measurement model validation

The measurement model was validated using confirmatory factor analysis (CFA) prior to structural model testing (Table 2). This CFA model was estimated independently of the subsequent structural models involving latent interaction terms. Confirmatory factor analysis results indicated a good model fit, with all key indices meeting or exceeding recommended thresholds (Hair et al., 2019; Hu & Bentler, 1999). Cronbach's alpha values for the latent constructs ranged from 0.77 to 0.90 (Cronbach, 1951). Composite reliability (CR) values ranged from 0.846 to 0.923, and average variance extracted (AVE) values ranged from 0.537 to 0.705, all above the accepted threshold of 0.50. These results demonstrated good internal consistency and convergent validity (Fornell & Larcker, 1981). Standardized factor loadings ranged from 0.51 to 0.87 (all significant), confirming that the items effectively represented their intended constructs. Furthermore, all inter-construct correlations were lower than the square root of the corresponding AVE, providing evidence of discriminant validity (Fornell & Larcker, 1981).

4.3. Structural model testing

A baseline structural model (M0), specifying the hypothesized main effects without interaction terms, was first estimated to examine the core relationships among AI-assisted speaking training affordances (AISTA), flow experience (FE), and willingness to communicate (WTC). This baseline model served as the reference model for subsequent moderated analyses.

The baseline structural model demonstrated an acceptable overall fit to the data ($\chi^2/df = 2.62$, CFI = 0.937, TLI = 0.927, SRMR = 0.052, RMSEA = 0.081, 90 % CI [.074, .089]). Although RMSEA was close to the upper bound of commonly recommended thresholds, the overall pattern of structural relationships was theoretically coherent and statistically stable, supporting the adequacy of the baseline model (Hair et al., 2019; Hu & Bentler, 1999).

The results of the three hypothesized main paths in the baseline model are summarized in Table 4. First, AI-assisted speaking training affordances (AISTA) were positively associated with flow experience (FE) (H1: $\beta = 0.692$, $p < .001$), indicating that learners

Table 2
Measurement model results (CFA) and latent correlations.

Construct	α	CR	AVE	FLR	AISTA	FE	WTC	LA
AISTA	0.826	0.872	0.537	[0.51, 0.79]	**0.733**			
FE	0.890	0.916	0.645	[0.69, 0.85]	0.724	**0.803**		
WTC	0.895	0.923	0.705	[0.80, 0.87]	0.509	0.690	**0.840**	
LA	0.773	0.846	0.525	[0.59, 0.82]	0.255	0.313	0.330	**0.725**

Note: α = Cronbach's alpha; CR = composite reliability; AVE = average variance extracted; FLR = factor loadings range. Values are derived from the CFA measurement model. Diagonal entries (bold) are square roots of AVE; off-diagonal entries are latent inter-construct correlations.

Table 3
Structural model results across nested steps (M0–M3).

Path (Std.)	M0	M1 (AISTA × LA → FE)	M2 (FE × LA → WTC)	M3 (Full)
FE ← AISTA	0.815***	0.815***	0.766***	0.763***
FE ← LA	0.156	0.156	0.159	0.170
FE ← AISTA×LA	–	0.252***	–	0.169**
WTC ← FE	1.196***	1.196***	0.731***	0.708***
WTC ← AISTA	−0.465	−0.465	−0.060	−0.035
WTC ← LA	0.095	0.095	0.124	0.150
WTC ← FE×LA	–	–	0.309***	0.137†
WTC ← AISTA×LA	–	–	–	0.183*
R^2(FE)	0.676	0.805	0.686	0.709
R^2(WTC)	0.671	0.774	0.724	0.744

Note.
Entries are standardized coefficients (β = std.all) estimated in lavaan.
†p < .10, *p < .05, **p < .01, ***p < .001.
Latent interaction terms were estimated using the product-indicator approach with double-mean centering (match = FALSE). Because interaction models rely on an expanded set of observed indicators, χ^2 difference tests against M0 are not emphasized.

Table 4
Structural path coefficients for the baseline model (M0).

Path	β	SE	t	p	Support
H1: AISTA → FE	0.692	0.044	15.75	<0.001	Yes
H2: FE → WTC	0.531	0.063	8.47	<0.001	Yes
H3: AISTA → WTC	0.074	0.061	1.23	0.221	No

Note. Standardized coefficients (β) are reported; SE = standard error. Estimates are obtained from the baseline latent-variable structural equation model (M0) without interaction terms, using maximum likelihood (ML) estimation.

who perceived and actively utilized AI-supported interactional resources were more likely to experience a state of immersion during speaking practice. Second, FE was positively related to willingness to communicate (WTC) (H2: β = 0.531, p < .001), suggesting that learners who experienced higher levels of flow tended to report stronger communicative motivation. In contrast, the direct path from AISTA to WTC was not statistically significant (H3: β = 0.074, p = .221), implying that the influence of technological affordances on communicative willingness was primarily indirect.

To further examine the robustness of this mediation structure, nested model comparisons were conducted by sequentially removing key structural components from the baseline model. Removing the direct path from AISTA to WTC did not result in a significant deterioration in model fit (Base vs. No-direct: $\Delta\chi^2 = 0.01$, p = .924), indicating that the direct effect of AISTA on WTC was not essential. By contrast, removing the mediating variable FE led to a substantial decline in model fit (Base vs. No-mediator: $\Delta\chi^2 = 78.41$, $\Delta df = 1$, p < .001), highlighting the central role of flow experience in linking AI-assisted speaking affordances to communicative outcomes.

4.4. Mediation analysis

To examine the mediating role of flow experience (FE) in the relationship between AI-assisted speaking training affordances (AISTA) and willingness to communicate (WTC), conditional indirect effects were estimated from the full moderated mediation model (M3) using BCa bootstrap procedures with 5000 resamples. Ninety-five percent confidence intervals (CIs) are reported in Table 5.

At low levels of learner agency (LA; −1 SD), the indirect effect of AISTA on WTC via FE did not reach statistical significance (effect = 0.502, 95 % CI [−0.066, 1.207]). This pattern suggests that when learners possess limited agency, AI-assisted affordances are less

Table 5
Conditional direct, indirect, and total effects of AISTA on WTC via FE at different levels of learner agency (M3).

Learner Agency Level	Indirect Effect (AISTA→FE→WTC)	95 % CI	Direct Effect (AISTA→WTC)	95 % CI	Total Effect	95 % CI
−1 SD (Low LA)	0.502	[−0.066, 1.207]	−0.381	[−1.046, 0.213]	0.121	[−0.270, 0.542]
Mean LA	1.067	[0.592, 1.776]	−0.069	[−0.720, 0.421]	0.998	[0.696, 1.311]
+1 SD (High LA)	1.819	[0.893, 3.091]	0.244	[−0.621, 0.972]	2.062	[1.489, 2.711]

Note. Conditional effects were estimated from the full moderated mediation model (M3) using maximum likelihood estimation and BCa bootstrap confidence intervals (5000 resamples). The model included both latent interaction terms (AISTA × LA and FE × LA). Effects are considered statistically significant when the 95 % confidence interval does not include zero.

consistently translated into enhanced communicative willingness through immersive learning experiences. Under this condition, neither the direct nor the total effect of AISTA on WTC was statistically reliable.

At the mean level of learner agency, the indirect effect of AISTA on WTC through FE was statistically significant (effect = 1.067, 95 % CI [0.592, 1.776]), while the direct effect remained non-significant. This finding indicates that under typical learning conditions, AI-assisted speaking affordances influence learners' willingness to communicate primarily through experiential mechanisms, with flow experience serving as the key mediating process.

At high levels of learner agency (+1 SD), the indirect effect further increased in magnitude and remained statistically significant (effect = 1.819, 95 % CI [0.893, 3.091]). Although the direct effect of AISTA on WTC was comparatively weak under this condition, the strengthened indirect pathway highlights the increasing efficiency with which highly agentic learners convert AI-assisted affordances into communicative motivation via immersive engagement.

Taken together, these results provide strong evidence for a conditional mediation pattern, in which flow experience functions as a central psychological mechanism linking AI-assisted speaking affordances to willingness to communicate, while learner agency shapes the strength and stability of this mechanism. Rather than exerting uniform effects across learners, AI-assisted affordances appear to facilitate communicative motivation in a differentiated manner, becoming increasingly effective as learners' agency enables deeper experiential engagement. This conditional pattern sets the stage for a closer examination of the moderating role of learner agency in the structural pathways, as illustrated in Fig. 2 and discussed in the following section.

4.5. *Moderation and conditional mediation analysis*

To further clarify how learner agency (LA) shapes the affordance–experience–communication mechanism, interaction terms in the full model (M3) were interpreted together with simple-slope visualizations (Fig. 2). High and low LA were probed at ±1 SD.

First, LA significantly moderated the AISTA → FE pathway (AISTA × LA → FE; Table 3), indicating that learners with higher agency were more effective in converting perceived AI-assisted affordances into flow-relevant engagement (Fig. 2a). Consistent with an ecological view of affordances, this pattern suggests that the same AI environment affords deeper immersion only when learners actively appropriate and mobilize available resources.

Second, the FE × LA interaction predicting WTC did not reach conventional significance in M3 (Table 3), although the simple slopes still showed a stronger positive FE → WTC association under high LA than under low LA (Fig. 2b). This suggests that agency may amplify the motivational consequences of flow, but this moderation should be interpreted cautiously given the statistical uncertainty of the interaction term.

Third, LA significantly moderated the AISTA → WTC pathway (AISTA × LA → WTC; Table 3). As shown in Fig. 2c, the direction of the direct affordance–WTC link differed across LA levels (positive under high LA and weakly negative under low LA). Importantly, this pattern is best interpreted as a conditional trend in the direct pathway, rather than decisive evidence that a direct effect consistently emerges for high-agency learners.

Taken together, the results indicate that LA primarily conditions the indirect pathway from AISTA to WTC by strengthening the affordance-to-flow link (and, descriptively, the flow-to-WTC link), yielding a pattern of conditional indirect effects (moderated mediation) rather than a uniform direct effect across learners.

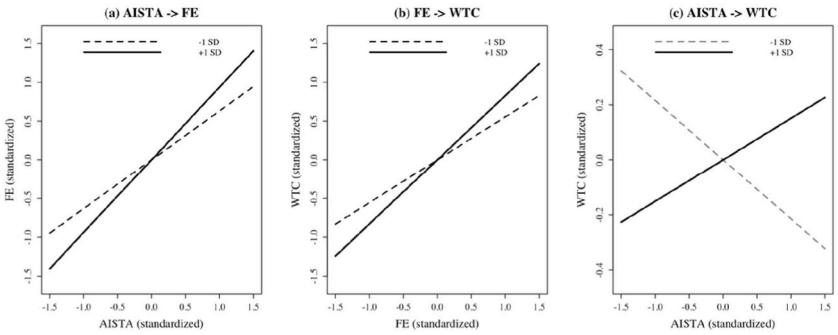


Fig. 2. Moderating effects of learner agency on structural paths

Note. Panels (a)–(c) visualize conditional associations implied by the full moderated mediation model (M3). Simple slopes are plotted at high (+1 SD) and low (−1 SD) levels of learner agency (LA). Panel (a) corresponds to the conditional AISTA → FE association; panel (b) corresponds to the conditional FE → WTC association; and panel (c) corresponds to the conditional direct AISTA → WTC association (reported for completeness and interpreted cautiously). All constructs were modeled as latent variables; slopes are based on standardized estimates from M3. Given the non-significant conditional direct effects in Table 5, panel (c) is presented as an illustrative pattern rather than confirmatory evidence of a reliable direct effect.

5. Discussion

5.1. The relationship between AISTA and WTC: the mediating role of FE

The findings indicate that the relationship between AISTA and WTC was primarily realized through the mediating role of FE. In the overall model, the direct effect of AISTA on WTC was not significant, whereas the indirect effect through the path "AISTA → FE → WTC" was significant (see Tables 4 and 5). This suggests that when learners engage with AI-assisted speaking resources, changes in their WTC are mainly achieved through the activation of flow. In other words, the interactivity, immediate feedback, and personalized support provided by AI environments are effectively transformed into higher levels of communicative motivation only when learners enter a state of immersion.

This finding underscores the pivotal role of FE in the "technology-psychology-behavior" chain. It is consistent with previous studies that have identified FE as a driving force of positive emotional experiences in language learning (Dewaele & Pavelescu, 2019; Dewaele et al., 2019; Mercer & MacIntyre, 2014; Shao et al., 2020). In this study, the mediating effect of FE was not only statistically significant but also of medium-to-large magnitude, aligning with the benchmarks suggested by Preacher and Kelley (2011). This demonstrates that FE is not a secondary psychological response but a core construct explaining the link between AISTA and WTC.

Taken together, the effect of AISTA on WTC was mainly achieved through the mediating pathway of FE. This finding reinforces the central role of FE in AI-supported language learning and provides new empirical evidence on how technological affordances are transformed into WTC through positive psychological experiences. Notably, the results resonate with prior work. For instance, Yang and Li (2024) found that challenge-skill balance and immediate feedback are critical conditions for fostering learner engagement in AI-assisted contexts, which is consistent with the mediating effect identified in this study. Importantly, the nonsignificant direct effect of AISTA on WTC (H3) should not be interpreted as a failure of AI-supported speaking environments. Rather, it provides theoretical evidence that technological affordances exert their influence on communicative motivation only when learners actively engage with and shape these affordances through their learning experience.

5.2. Multi-level moderation of LA: Why the key to Human-AI interaction lies in the "human"

After confirming the mediating role of FE, this study further examined the moderating role of LA. The results revealed that LA exerted significant effects on multiple pathways in the model, including "AISTA → FE," "FE → WTC," and the direct path "AISTA → WTC." Overall, learners with higher levels of LA exhibited a more robust and stable pattern of experiential engagement in AI-assisted speaking training. This finding not only supports the interaction hypotheses but also aligns with the ecological linguistic view that environmental resources must be actively mobilized by agents in order to translate into learning outcomes (Kordt, 2016; Van Lier, 2004). This indicates that the true driving force behind AI-enabled learning does not lie in the algorithm or platform itself, but rather in learners' ability to perceive, select, and reconstruct technological resources—that is, in their agentic empowerment rather than passive technologization.

First, on the "AISTA → FE" path, LA significantly moderated the relationship (see Table 3, Fig. 2a). Learners with high LA were better able to draw on the multidimensional resources of AISTA, creating a challenge-skill balance and thereby entering a state of flow more easily. In contrast, low-LA learners tended to fall into "affordance blindness," failing to consistently achieve flow due to their lack of initiative. This finding supports van Lier's (2004) argument that affordances exist only when they are perceived and activated by learners. From an educational psychology standpoint, this divergence essentially reflects variations in perceived control and task value (Pekrun, 2006). High-agency learners, through proactive engagement with AI-mediated affordances, cultivate a heightened sense of controllability and meaningfulness, which in turn fosters sustained engagement and a more enduring state of enjoyment.

Second, on the "FE → WTC" path, learner agency appeared to condition the strength of the association, as reflected in the simple slope patterns shown in Fig. 2b. High-LA learners were more likely to translate the enjoyment and confidence they experienced during AI practice into stronger links between FE and WTC, actively seeking communicative opportunities both inside and outside the classroom. By contrast, low-LA learners might experience positive emotions temporarily during practice but often lacked sustained agency due to "learned passivity" (Seyri & Rezaee, 2024), which weakened the long-term relationship between FE and WTC. Previous research has also noted that not all positive emotional experiences translate into communicative behavior, with individual differences in LA serving as a critical conditional factor (Lee & Chiu, 2024; Shao et al., 2020). This finding further reveals that the activation of flow does not inherently lead to communicative action; rather, its effectiveness depends on whether learners possess an agentic mechanism that enables them to convert emotional experience into communicative motivation—that is, a self-regulatory process linking emotion and action within human-AI interaction.

Finally, the moderating effect of LA on the direct path "AISTA → WTC" proved particularly noteworthy. While this path was nonsignificant in the overall model (see Table 4), the simple slope patterns suggest that a positive AISTA → WTC association was more likely to emerge under high-LA conditions, although the conditional direct effects themselves did not reach statistical significance (see Table 5). Low-LA learners, however, showed slopes close to zero or even slightly negative. This indicates that AI environments with diverse functions and high degrees of freedom are not necessarily "friendly" to all learners. For those with low LA, the complexity of interactive options and feedback may trigger cognitive overload or choice overload, undermining their sense of control and self-efficacy, thereby suppressing WTC. This phenomenon echoes concerns about a "technological inequality effect," where learners with high and low levels of LA experience markedly different benefits from the same AI environment (Liu et al., 2025). This suggests that the proliferation of AI technology does not automatically lead to equity; rather, it may amplify learning disparities through differences in learner agency. Designers and educators therefore need to recognize that the friendliness of AI-assisted environments

ultimately depends on whether they can provide differentiated scaffolding and adaptive pacing that accommodate learners with varying levels of agency.

In sum, LA not only determines whether technological affordances can be effectively utilized in the learning process but also shapes the strength of the connection between positive experiences and genuine WTC. This finding reinforces the dynamic interpretation of the technology-psychology-behavior chain. In AI-empowered learning environments, the root of learning variation lies not in the technology itself but in whether learners actively mobilize and exercise their own agency (Guo & Li, 2024). Therefore, the future of human-AI collaboration should not be understood as AI replacing humans, but rather as AI amplifying humans—its true value lies in activating, rather than substituting, learners' will to learn and their capacity for self-growth. This human-centered perspective offers both ethical and pedagogical implications for AI-empowered education and lays a foundational framework for future research on human-centered AI learning design.

6. Conclusion, limitations, and future directions

The findings of this study confirmed the dynamic assumptions proposed in the "Affordance-Experience-Motivation" chain model, demonstrating that the effectiveness of AI-assisted speaking training is not automatically determined by technology itself but depends on how learners perceive, activate, and transform ecological affordances (AISTA) into meaningful learning actions. Grounded in Affordance Theory and Flow Theory within the framework of WTC, this study systematically examined the interrelationships among AISTA, FE, LA, and WTC among Chinese EFL university students.

The results revealed that AISTA was significantly correlated with FE, and FE showed a significant positive relationship with WTC. Learners who actively utilized AI-provided feedback and contextual resources were more likely to experience a state of immersion characterized by a challenge-skill balance, which in turn fostered stronger communicative motivation. Moreover, FE served as a mediator between AISTA and WTC, suggesting that the pedagogical value of AI environments is primarily realized through learners' psychological experiences. Notably, while the direct effect of AISTA on WTC was non-significant in the overall model (and remained statistically non-significant across probed LA levels), the indirect effect via flow strengthened as learner agency increased, indicating a pattern of conditional indirect effects (moderated mediation). This highlights learner agency as the pivotal link that connects technological potential with communicative behavior.

Further analysis showed that LA exerted significant moderating effects on all three core pathways: learners with high agency were more capable of activating multidimensional affordances, thereby strengthening the association between FE and WTC, and transferring AI-mediated experiences directly into real communicative contexts. This finding substantiates the principle that the educational value of AI depends on human agency, emphasizing the central role of individual differences in AI-empowered learning. It also provides empirical evidence for the development of learner-centered human-AI collaborative models in future intelligent learning environments.

Despite these contributions, the study has several limitations that call for further research. First, the scope of the sample limits the external validity of the findings. The participants were Chinese EFL undergraduates, a group with relatively homogeneous cultural and demographic characteristics. Caution is therefore warranted when generalizing the results to other populations and contexts. Future research should test the model across different cultural and learner groups, using measurement invariance testing and multilevel modeling to enhance generalizability and robustness (Vandenberg & Lance, 2000).

Second, the reliance on a single data source may have introduced common method bias. Although multiple procedural and statistical controls were employed in this study, the use of self-report questionnaires cannot fully eliminate the risk of confounding or underestimating effects (Podsakoff et al., 2003). Future research should adopt multi-source and mixed-method designs. For instance, classroom recordings or ethnographic observations could be used to capture behavioral evidence, AI system logs could be extracted to quantify interaction frequency and feedback patterns, and wearable sensors could monitor physiological indicators such as heart rate variability and skin conductance. These multimodal data sources would allow for triangulation of psychological and behavioral constructs, thereby reducing common method bias and enhancing reliability and explanatory power (Liu et al., 2025).

Third, the cross-sectional design limits causal inference. As this study employed a cross-sectional survey design, it was unable to capture causal directionality or dynamic processes. The relationships among AISTA, FE, WTC, and LA may not be static or linear, but instead fluctuate over time and across contexts. Both FE and WTC are state-like variables, often shaped by immediate classroom conditions (Lee & Chiu, 2024). Future studies could employ longitudinal or experimental designs to uncover causal pathways and temporal dynamics. On the one hand, longitudinal tracking or experience sampling could be used to observe the developmental trajectories of AISTA, FE, and WTC over a semester or longer, thereby revealing the cumulative effects of sustained AI-assisted speaking practice on psychological and behavioral outcomes (Lee & Chiu, 2024; Mercer & MacIntyre, 2014). On the other hand, experimental manipulations of AISTA or LA (e.g., by providing different intensities of AI practice or implementing pedagogical interventions to foster LA) could help validate the causal links and boundary conditions of the core pathways, thereby strengthening causal explanations (Hwang et al., 2025; Reeve & Tseng, 2011).

Fourth, the restricted range of variables limited the comprehensiveness of the theoretical framework. WTC is a complex construct influenced by multiple factors. For instance, motivation, language anxiety, and teacher support have all been shown to play important roles (Khajavy et al., 2018; Liu & Jackson, 2008), yet these factors were not included in the present model. Future research could expand the framework by incorporating additional variables. For example, it would be useful to examine how the interplay of positive emotions (e.g., enjoyment, engagement) and negative emotions (e.g., anxiety, frustration) affects WTC, or to consider environmental support factors such as teacher feedback styles and peer collaboration, and how these interact with the affordances provided by AI. By broadening the scope of variables, researchers can develop a more comprehensive explanatory framework, shifting from a

single-pathway model to an integrated ecological system that better captures the factors and mechanisms shaping WTC in AI-enabled environments.

Fifth, the reliance on a single platform may have limited the external validity of the findings. This study used *Tencent Yuanbao*, a relatively specific AI-based speaking practice application, whose functional features may have influenced the results. For instance, different AI systems vary in speech recognition accuracy, feedback modes, and conversational styles (Lo, 2025), and these differences may shape learners' experiences and outcomes. Thus, the generalizability of the present findings to other AI platforms remains uncertain. Future research should validate and compare the model across multiple platforms and tools. For example, applying the present measures and model to open-domain dialogue systems such as ChatGPT or other mainstream speaking practice applications could help determine whether the relationships among AISTA, FE, LA, and WTC hold consistently. If differences emerge across platforms, further analysis should examine how specific technological features contribute to those variations. Expanding data collection to broader samples and environments will improve the external validity of the findings and provide targeted insights for optimizing the educational applications of different AI tools.

In sum, the limitations of this study highlight that AI technology itself is not a decisive factor in educational effectiveness. Learner individual differences, cultural contexts, research methodologies, and technological choices can all profoundly shape the relationships among AISTA, FE, and WTC. Future studies should advance theoretical models and pedagogical practices within frameworks that are cross-cultural, multi-method, dynamic, and multi-platform. Such efforts will contribute to promoting both the equity and effectiveness of AI-enabled language learning.

Ethics approval

This study was reviewed and approved by the Academic Ethics and Moral Oversight Committee (Ethics Review Section) of Hubei University of Science and Technology (Approval No. 202503018). All participants provided informed consent prior to data collection, and the study was conducted in accordance with the Declaration of Helsinki.

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Conflict of interest statement

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.system.2026.103986.

References

Albedah, F. (2025). Artificial intelligence in language education: A systematic review of multilingual applications, large language models, and emerging challenges. *Language Teaching Research Quarterly, 49*, 247–268. https://doi.org/10.32038/ltrq.2025.49.13

Alrabai, F. (2022). The role of mixed emotions in language learning and teaching: A positive psychology teacher intervention. *System, 107*, Article 102821. https://doi.org/10.1016/j.system.2022.102821

Cronbach, L. J. (1951). Coefficient alpha and the internal structure of tests. *Psychometrika, 16*, 297–334.

Csikszentmihalyi, M. (1990). *Flow: The psychology of optimal experience*. HarperCollins.

Dewaele, J.-M., Chen, X., Padilla, A. M., & Lake, J. (2019). The flowering of positive psychology in foreign language teaching and acquisition research. *Frontiers in Psychology, 10*, 2128. https://doi.org/10.3389/fpsyg.2019.02128

Dewaele, J.-M., & Dewaele, L. (2018). Learner-internal and learner-external predictors of willingness to communicate in the FL classroom. *Journal of the European Second Language Association, 2*(1), 24–37. https://doi.org/10.22599/jesla.37

Dewaele, J.-M., & Pavelescu, L. M. (2019). The relationship between incommensurable emotions and willingness to communicate in English as a foreign language: A multiple case study. *Innovation in Language Learning and Teaching, 13*(3), 310–324. https://doi.org/10.1080/17501229.2019.1675667

Egbert, J. (2004). A study of flow theory in the foreign language classroom. *Canadian Modern Language Review, 60*(5), 549–586. https://doi.org/10.3138/cmlr.60.5.549

Engeser, S., & Rheinberg, F. (2008). Flow, performance and moderators of challenge–skill balance. *Motivation and Emotion, 32*(3), 158–172. https://doi.org/10.1007/s11031-008-9102-4

Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research, 18*(1), 39–50.
Gibson, J. J. (1979). *The ecological approach to visual perception*. Houghton Mifflin.
Guo, K., & Li, D. (2024). Understanding EFL students' use of self-made AI chatbots as personalized writing assistance tools: A mixed methods study. *System, 124*, Article 103362. https://doi.org/10.1016/j.system.2024.103362
Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019). *Multivariate data analysis* (8th ed.). Cengage Learning.
Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal, 6*(1), 1–55. https://doi.org/10.1080/10705519909540118
Hwang, G.-J., Darmawansah, D., Lin, C.-J., & Febiyani, F. (2025). An artificial intelligence-supported GFCA learning model to enhance L2 students' role-play performance, English speaking and interaction mindset. *Educational Technology Research & Development, 73*, 1451–1479. https://doi.org/10.1007/s11423-025-10453-6
Kearney, M., Burden, K., & Rai, T. (2019). The iPAC scale: A survey to measure distinctive mobile pedagogies. *TechTrends, 63*(6), 685–696. https://doi.org/10.1007/s11528-019-00414-1
Khajavy, G. H., MacIntyre, P. D., & Barabadi, E. (2018). Role of the emotions and classroom environment in willingness to communicate: Applying doubly latent multilevel analysis in second language acquisition research. *Studies in Second Language Acquisition, 40*(3), 605–624. https://doi.org/10.1017/S0272263117000304
Kim, A., & Su, Y. (2024). How implementing an AI chatbot impacts learners' willingness to communicate in Korean. *System, 122*, Article 103256. https://doi.org/10.1016/j.system.2024.103256
Kirkpatrick, R., Vafadar, H., & Mohebbi, H. (2024). A scoping review of willingness to communicate in language education: Premises to doubt, lessons to learn, and future research questions to examine. *Language Testing in Asia, 14*. https://doi.org/10.1186/s40468-024-00284-2. Article 9.
Kordt, B. (2016). Affordance theory and multiple language learning and teaching. *International Journal of Multilingualism, 15*(2), 1–14. https://doi.org/10.1080/14790718.2016.1223081
Lai, W. Y. W., & Lee, J. S. (2024). A systematic review of conversational AI tools in ELT: Publication trends, tools, research methods, learning outcomes, and antecedents. *Computers and Education: Artificial Intelligence, 7*, Article 100291. https://doi.org/10.1016/j.caeai.2024.100291
Lee, J. S., & Chiu, M. M. (2024). Modelling trait and state willingness to communicate in a second language: An experience sampling approach. *Studies in Second Language Learning and Teaching, 14*(3), 483–514. https://doi.org/10.14746/ssllt.37541
Lee, H., & Lee, J. H. (2024). The effects of AI-guided individualized language learning: A meta-analysis. *Language, Learning and Technology, 28*(2), 134–162. https://doi.org/10.64152/10125/73575
Liu, G. L., Darvin, R., & Ma, C. (2025). Exploring AI-mediated informal digital learning of English (AI-IDLE): A mixed-method investigation of Chinese EFL learners' AI adoption and experiences. *Computer Assisted Language Learning, 38*(7), 1632–1660. https://doi.org/10.1080/09588221.2024.2310288
Liu, M., & Jackson, J. (2008). An exploration of Chinese EFL learners' unwillingness to communicate and foreign language anxiety. *The Modern Language Journal, 92*(1), 71–86. https://doi.org/10.1111/j.1540-4781.2008.00687.x
Lo, A. W. T. (2025). The educational affordances and challenges of generative AI in Global Englishes oriented materials development and implementation: A critical ecological perspective. *System, 130*, Article 103610. https://doi.org/10.1016/j.system.2025.103610
McMillan, S. J., & Hwang, J.-S. (2002). Measures of perceived interactivity: An exploration of the role of direction of communication, user control, and time in shaping perceptions of interactivity. *Journal of Advertising, 31*(3), 29–42. https://doi.org/10.1080/00913367.2002.10673674
Mercer, S., & MacIntyre, P. D. (2014). Introducing positive psychology to SLA. *Studies in Second Language Learning and Teaching, 4*(2), 153–172. https://doi.org/10.14746/ssllt.2014.4.2.2
Pekrun, R. (2006). The control-value theory of achievement emotions: Assumptions, corollaries, and implications. *Educational Psychology Review, 18*(4), 315–341. https://doi.org/10.1007/s10648-006-9029-9
Peng, J.-E., & Liang, W. (2025). Willingness to communicate with artificial intelligence (AI)? Insights from tracking EFL learners' perceived acceptance and chat output. *Computer Assisted Language Learning*. https://doi.org/10.1080/09588221.2025.2486147
Peng, J.-E., & Woodrow, L. (2010). Willingness to communicate in English: A model in the Chinese EFL classroom context. *Language Learning, 60*(4), 834–876. https://doi.org/10.1111/j.1467-9922.2010.00576.x
Podsakoff, P. M., MacKenzie, S. B., Lee, J.-Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology, 88*(5), 879–903. https://doi.org/10.1037/0021-9010.88.5.879
Preacher, K. J., & Kelley, K. (2011). Effect size measures for mediation models: Quantitative strategies for communicating indirect effects. *Psychological Methods, 16*(2), 93–115. https://doi.org/10.1037/a0022658
Reeve, J., & Tseng, C.-M. (2011). Agency as a fourth aspect of students' engagement during learning activities. *Contemporary Educational Psychology, 36*(4), 257–267. https://doi.org/10.1016/j.cedpsych.2011.05.002
Rheinberg, F., Vollmeyer, R., & Engeser, S. (2003). *The Flow State Scale (FSS)*. University of Jena, Department of Psychology.
Sato, R. (2025). Japanese EFL speakers' fluctuation of willingness to communicate: Its interplay and association with feedback. *Language Teaching Research Quarterly, 48*, 69–83. https://doi.org/10.32038/ltrq.2025.48.05
Scrivener, J. (2011). *Learning teaching: The essential guide to English language teaching* (3rd ed.). Education: Macmillan.
Seyri, H., & Rezaee, A. A. (2024). An autonomy-oriented response to EAP students' learned helplessness in online classes. *Current Psychology, 43*(9), 7887–7898. https://doi.org/10.1007/s12144-023-04981-8
Shao, K., Pekrun, R., Marsh, H. W., & Loderer, K. (2020). Control-value appraisals, achievement emotions, and foreign language performance: A latent interaction analysis. *Learning and Instruction, 69*, Article 101356. https://doi.org/10.1016/j.learninstruc.2020.101356
Song, Y. (2021). Affordances of a mobile learner-generated tool for pupils' English as a second language vocabulary learning: An ecological perspective. *British Journal of Educational Technology, 52*(5), 1823–1838. https://doi.org/10.1111/bjet.13076
Tai, T.-Y., & Chen, H. H.-J. (2024). Improving elementary EFL speaking skills with generative AI chatbots: Exploring individual and paired interactions. *Computers & Education, 220*, Article 105112. https://doi.org/10.1016/j.compedu.2024.105112
Tarp, G. (2025). English as a lingua franca and its impact on willingness to communicate in a second language. *Language Teaching Research Quarterly, 48*, 125–138. https://doi.org/10.32038/ltrq.2025.48.08
Van Lier, L. (2004). *The ecology and semiotics of language learning: A sociocultural perspective*. Springer. https://doi.org/10.1007/1-4020-7912-5
Vandenberg, R. J., & Lance, C. E. (2000). A review and synthesis of the measurement invariance literature: Suggestions, practices, and recommendations for organizational research. *Organizational Research Methods, 3*(1), 4–70. https://doi.org/10.1177/109442810031002
Yang, L., & Li, R. (2024). ChatGPT for L2 learning: Current status and implications. *System, 124*, Article 103351. https://doi.org/10.1016/j.system.2024.103351
Zuniga, M. (2023). The correlates of flow in the L2 classroom: Linking basic L2 task features to learner flow experiences. *The Modern Language Journal, 107*(3), 650–668. https://doi.org/10.1111/modl.12865